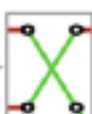






► $P_t^{(n,n+1)} \in \{0, 1\}$ indicated if the n -th swap is proposed

$t = 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9$
 $O = P_{\text{odd}}$
 $E = P_{\text{even}}$
 $E O E O E O E O E O$
 7


Proposal
indicators

$$P_7^{(5,6)} = 0$$

$$P_8^{(3,4)} = 1$$

$$P_8^{(1,2)} = 1$$



$$\mathbf{K}_{\text{comm}} = \prod_{n=1}^N K^{(n,n+1)} P_t^{(n,n+1)}$$

- These are the largest collection of swap moves in parallel



A specialist and even swaps

$$\mathbf{K}_{\text{comm}} \in \{\mathbf{K}_{\text{odd}}, \mathbf{K}_{\text{even}}\}, \quad \mathbf{K}_{\text{odd}} = \prod_{n \text{ odd}} \mathbf{K}^{(n, n+1)}, \quad \mathbf{K}_{\text{even}} = \prod_{n \text{ even}} \mathbf{K}^{(n, n+1)}$$



Note that in general, the order matters

► Order does not matter!

COMMUNICATION

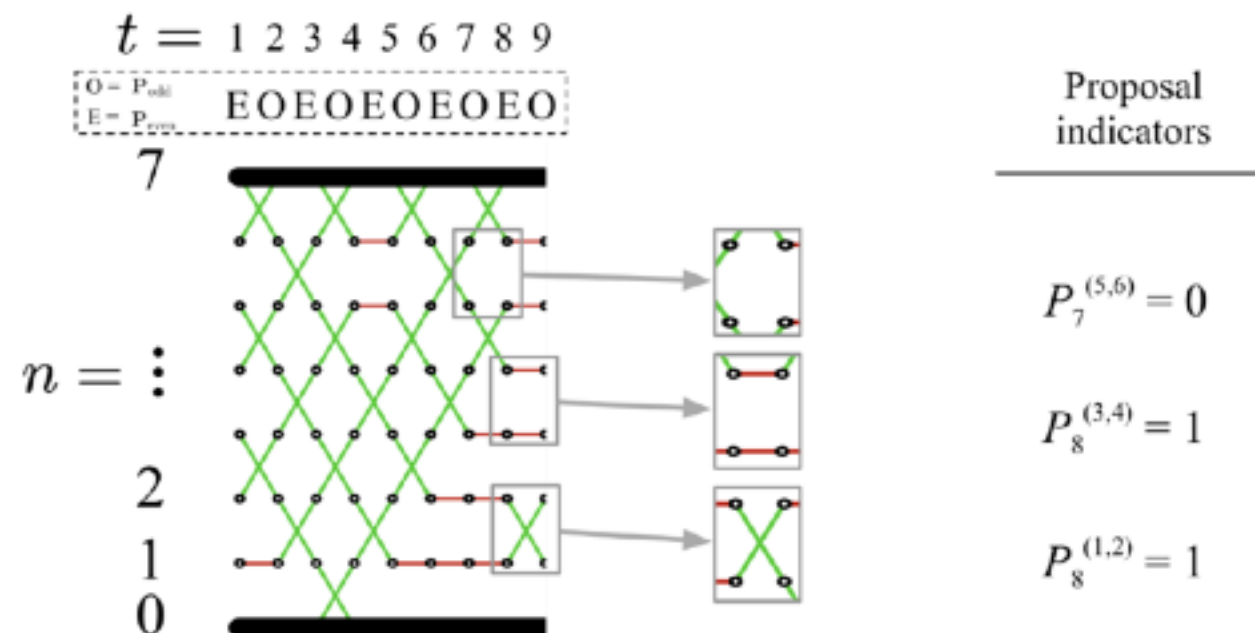
► Communication move

$$\mathbf{K}_{\text{comm}} = \prod_{n=1}^N K^{(n,n+1)} P_t^{(n,n+1)}$$

- $P_t^{(n,n+1)} \in \{0,1\}$ indicated if the n -th swap is proposed
- Note that in general, the order matters
- A special is odd and even swaps

$$\mathbf{K}_{\text{comm}} \in \{\mathbf{K}_{\text{odd}}, \mathbf{K}_{\text{even}}\}, \quad \mathbf{K}_{\text{odd}} = \prod_{n \text{ odd}} \mathbf{K}^{(n,n+1)}, \quad \mathbf{K}_{\text{even}} = \prod_{n \text{ even}} \mathbf{K}^{(n,n+1)}$$

- These are the largest collection of swap moves in parallel
- Order does not matter!



PARALLEL TEMPERING